FLORENT TARIOLLE

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Seeking a six-month research internship abroad in world models, starting April 2027.

Education

INSA Rouen Normandy 2023 – 2027
Engineering degree, Computer Science and Artificial Intelligence; expected 2027

University of Rouen Normandy 2026 – 2027
Master's degree, Machine Learning for Artificial Intelligence; expected 2027

Lycée Dupuy de Lôme 2021 – 2023
Two-year intensive preparatory programme in mathematics and physics (CPGE MP)

Publications

DashVMC: Real-Time Discrete World Model Control in Geometry Dash Sep 2026
F. Tariolle and F. Yger. PTA workshop, NeurIPS 2026 — accepted paper; poster.

arXiv: 2609.40003 · **Demo** · **Code**

- Developed a world-model agent for live Geometry Dash control from pixels, refining a behaviour-cloned policy with PPO entirely in frozen-model rollouts.
- Trained offline on approximately two hours of gameplay; improved survival over cloning on three official levels and a held-out layout, with 60-FPS control on an RTX 2060.

Opportunistic Target Selection: Early Directional Commitment for Query-Efficient Black-Box Adversarial Attacks May 2026
F. Tariolle and F. Yger. CAp 2026 (Conférence sur l'Apprentissage automatique) — accepted paper; poster.

arXiv: 2605.25663 · **Code** · **Poster**

- Developed a wrapper for score-based black-box adversarial attacks that uses early class-confidence signals to select a target, reducing class drift without gradient access.
- Evaluated 4,500 ImageNet attack runs across five classifiers. On ResNet-50, improved attack success by up to 27 percentage points and reduced censored-mean attack iterations by 43%.

Research Projects and Awards

Finalist, Hack the World(s) Hackathon Jun 2026
Geometry-aware joint-embedding EEG · 24-hour JEPA and world-model challenge

Code · **Presentation**

Selected as one of 100 participants from 650+ applicants for a hackathon sponsored by Yann LeCun. Our team finished among the top five of 25 teams.

Built a model that learns from unlabeled EEG recordings. Explored whether training that respects the curved geometry of covariance matrices leads to more structured representations.

Experience

InterDigital

Generative AI Consultant Oct 2026 – Present

Freelance · France, remote

Advising InterDigital's R&D teams on generative AI workflows, plugins, and skills, building on tools I initiated during my research internship. Helping teams develop and maintain these tools independently through practical guidance on AI-assisted development, reliability, and security.

Deep Learning Research Intern May – Sep 2026

Rennes, France

Researched learned decoder-side filter selection as an alternative to neural in-loop filtering for VVC (H.266) video decoding, with the aim of reducing compression and quantization artifacts. Contributed to more than 10 patent applications arising from this research.

Enedis

Machine Learning Intern Jan 2026 – Present

Part-time capstone project, 25 hours/week · Rouen, France

Building a digital twin of Enedis's distribution network by mapping synthetic customer load profiles to distribution substations and linking these to primary substations. Combining DeepCourboGen-generated curves with measured substation loads, and developing spatial and temporal adjustments for more realistic local profiles.

Paused May–September 2026 during the research internship at InterDigital.

Open-Source Projects

VisualTorch Jul 2026 – Present

Maintainer and MCP Integration Lead · Official PyTorch Ecosystem project

Maintaining a library for visualizing PyTorch neural-network architectures. Designed and implemented its MCP integration to generate editable diagrams from model definitions, making architecture inspection available through agent tooling. The project has more than 300 GitHub stars.

Rose — Real-time customization for League of Legends Sep – Dec 2025

Co-founder and Lead Developer · Open source

Co-founded and co-developed an open-source customization tool; remain the project's top contributor. Built core application components spanning the Python backend, in-client JavaScript plugins, and their real-time WebSocket bridge.

The project has since reached 30,000 daily active users and 90,000 monthly active users, with 700+ GitHub stars and 20+ contributors.

Technical Skills

Programming: Python, C++
Frameworks: PyTorch
Research: Self-supervised learning, world models, model-based reinforcement learning and planning
Languages: French (native), English (TOEIC 970/990), Spanish (intermediate)

Interests

League of Legends (top 0.1% of ranked players; two LAN wins), bouldering, and strength training.

References

Florian Yger (INSA Rouen Normandy)
Research supervisor and co-author
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Frédéric Lefèbvre (InterDigital)
Internship and consulting supervisor
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